

In the staggered-fermion formulation, we consider finite one-dimensional lattice of N sites with fermionic variables χ_n , normalized as $\{\chi_n^\dagger, \chi_m\} = \delta_{nm}$, at each lattice site and N is even. Here, the sites are indexed as $n = 0, 1, \dots, N-1$ with discretized lattice spacing a . We consider a system with open boundary conditions.

Electric field operator,

$$L_n - L_{n-1} = \chi_n^\dagger \chi_n - \frac{1 + (-1)^n}{2} + Q_n^+(t) - Q_n^-(t), \quad (1)$$

with $L_{-1} = 0$ and $L_{N-1} = 0$ in a state constrained to $N/2$ fermions and zero total charge. Two moving charges with trajectories, $x = \pm vt$ are smeared across neighboring sites as

$$Q_n^\pm(t) = Q \cdot \max(1 - |n_0 \pm vt/a - n|, 0). \quad (2)$$

Components of energy-momentum tensor are discretized as

$$\begin{aligned} T_n^{00} &= \frac{i}{4a^2}(K_n + K_{n-1}) + \frac{g^2}{4}(L_n^2 + L_{n-1}^2) + \frac{m}{a}(-1)^n \chi_n^\dagger \chi_n, \\ T_n^{11} &= \frac{i}{4a^2}(K_n + K_{n-1}) - \frac{g^2}{4}(L_n^2 + L_{n-1}^2), \\ T_n^{01} &= \frac{i}{4a^2}(\chi_{n+1}^\dagger \chi_{n-1} - \chi_{n-1}^\dagger \chi_{n+1}), \end{aligned} \quad (3)$$

where the kinetic term component $K_n = \chi_{n+1}^\dagger \chi_n - \chi_n^\dagger \chi_{n+1}$ and $K_{-1} = K_{N-1} = 0$. The Hamiltonian reads

$$\mathbb{H} = a \sum_{n=0}^{N-1} T_n^{00} = \frac{i}{2a} \sum_{n=0}^{N-2} (\chi_{n+1}^\dagger \chi_n - \chi_n^\dagger \chi_{n+1}) + m \sum_{n=0}^{N-1} (-1)^n \chi_n^\dagger \chi_n + \frac{g^2 a}{2} \sum_{n=0}^{N-1} L_n^2 \quad (4)$$

Centering in the middle of the lattice $n_0 = (N-1)/2$, the boost operator reads

$$\begin{aligned} \mathbb{K} &= -a^2 \sum_{n=0}^{N-1} (n - n_0) T_n^{00} \\ &= \frac{-i}{2} \sum_{n=0}^{N-1} \left(n + \frac{1}{2} - n_0\right) (\chi_{n+1}^\dagger \chi_n - \chi_n^\dagger \chi_{n+1}) - am \sum_{n=0}^{N-1} (-1)^n (n - n_0) \chi_n^\dagger \chi_n - \frac{g^2 a^2}{2} \sum_{n=0}^{N-1} \left(n + \frac{1}{2} - n_0\right) L_n^2 \end{aligned} \quad (5)$$

We define current operators and chiral condensate density for pairs or sites

$$\begin{aligned} j_i^0 &= \frac{1}{2a} (\chi_{2i}^\dagger \chi_{2i} + \chi_{2i+1}^\dagger \chi_{2i+1}), \\ j_i^1 &= \frac{1}{2a} (\chi_{2i}^\dagger \chi_{2i+1} + \chi_{2i+1}^\dagger \chi_{2i}), \\ \nu_i &= \frac{1}{2a} (\chi_{2i}^\dagger \chi_{2i} - \chi_{2i+1}^\dagger \chi_{2i+1}), \end{aligned} \quad (6)$$

We boost the first excited state $|P = 0\rangle$ as

$$|P\rangle = e^{-i\chi\mathbb{K}}|P = 0\rangle$$

where χ follows as

$$P = E \frac{e^{2\chi} - 1}{e^{2\chi} + 1},$$

and E is the energy above the ground state.